

AVESTA · EGYPT RESIDENTIAL

Home Tier Sizing

The equations behind the tool, from the customer's bill to battery size, savings and payback

Worked example on every slide: a monthly bill of 2,000 EGP

OVERVIEW

Five steps from a bill to a recommendation



Example: 2,000 EGP → 851 kWh per month → **Solar only** 3 kWp, or **Solar + battery** 6.5 kWp with 21.12 kWh

INPUT DATA

Residential tariff tiers (August 2026 bill)

Tier	Monthly kWh	EGP / kWh	How it bills
1	0 – 50	0.68	Progressive
2	51 – 100	0.87	Progressive
3	101 – 200	1.00	Progressive; above 100 kWh the first 200 all bill at 1.00
4	201 – 350	1.73	Progressive
5	351 – 650	2.18	Progressive
6	651 – 1000	2.35	Flat on ALL kWh
7	> 1000	2.89	Flat on ALL kWh

Fees and future tariff increases are ignored. Rates are editable in the tool.

STEP 2 • TARIFF

Monthly bill from consumption M (kWh)

Consumption	Bill (EGP)
M ≤ 50	0.68 × M
50 < M ≤ 100	34 + 0.87 × (M − 50)
100 < M ≤ 650	1.00 × min(M, 200) + 1.73 × max(0, min(M, 350) − 200) + 2.18 × max(0, M − 350)
650 < M ≤ 1000	2.35 × M
M > 1000	2.89 × M

Check values	
M (kWh)	Bill (EGP)
90	68.8
300	373.0
500	786.5
800	1,880.0
1,200	3,468.0

First, the bill at each tier boundary

Consumption	Calculation	Bill (EGP)
50 kWh	50×0.68	34
100 kWh	$34 + 50 \times 0.87$	77.5
200 kWh	200×1.00	200
350 kWh	$200 + 150 \times 1.73$	459.5
650 kWh	$459.5 + 300 \times 2.18$	1,113.5
1000 kWh	1000×2.35	2,350

Method

1. Find the tier: compare the bill with these boundary bills.

2. Subtract the cost of the tiers below it.

3. Divide what is left by that tier's rate.

Above 100 kWh the first 200 kWh all bill at 1.00; above 650 kWh all consumption bills at one flat rate.

STEP 2 • BILL → KWH

Consumption M from the bill B

If the bill B is	Tier	Consumption M (kWh)
up to 34	1	$B \div 0.68$
34 – 77.5	2	$50 + (B - 34) \div 0.87$
101 – 200	3	$B \div 1.00$
200 – 459.5	4	$200 + (B - 200) \div 1.73$
459.5 – 1,113.5	5	$350 + (B - 459.5) \div 2.18$
1,527.5 – 2,350	6	$B \div 2.35$
2,890 and above	7	$B \div 2.89$

Bills between 77.5–101, 1,113.5–1,527.5 and 2,350–2,890 have no exact consumption (see the jumps slide).

STEP 2 • BILL → KWH

Four worked examples

300 EGP

Between 200 and 459.5 → Tier 4

$$200 + (300 - 200) \div 1.73 = \mathbf{257.8 \text{ kWh}}$$

1,000 EGP

Between 459.5 and 1,113.5 → Tier 5

$$350 + (1,000 - 459.5) \div 2.18 = \mathbf{597.9 \text{ kWh}}$$

2,000 EGP

Between 1,527.5 and 2,350 → Tier 6

$$2,000 \div 2.35 = \mathbf{851.1 \text{ kWh}}$$

4,000 EGP

Above 2,890 → Tier 7

$$4,000 \div 2.89 = \mathbf{1,384.1 \text{ kWh}}$$

STEP 2 · BILL → KWH

How the tool computes it, and the bill jumps

M = largest m where Bill(m) ≤ B

The tool gets the same result by bisection (60 halvings, rounded to 0.1 kWh), so it stays correct if the tier rates are edited.

1,000 EGP → 597.9 kWh
2,000 EGP → 851.1 kWh

Bills inside a jump

At	Bill jumps (EGP)	Mapped to
100 kWh	77.5 → 101	100 kWh
650 kWh	1,113.5 → 1,527.5	650 kWh
1000 kWh	2,350 → 2,890	1000 kWh

No consumption gives these bills exactly (fees are the usual cause), so the tool uses the threshold. Better: take the kWh printed on the bill.

STEP 2 · DAILY ENERGY

Daily consumption and its day / night split

$$E = M \div 30$$

$$N = E \times \text{night share (default 55\%)}$$

$$D = E - N$$

28.37

E · kWh per day (851.1 ÷ 30)

15.60

N · kWh at night

12.77

D · kWh in daytime

STEP 3 · SOLAR PV

Solar array size P (kWp)

Daily generation $G = P \times Y$, where $Y = 4.8$ kWh per kWp per day

Solar only

$$P = D \div Y$$

$$12.77 \div 4.8 = 2.66 \rightarrow \mathbf{3.0 \text{ kWp}}$$

Without a battery, solar beyond the daytime load is lost.

Solar + battery

$$P = (D + \min(N, U) \div \eta) \div Y$$

$$(12.77 + 15.60 \div 0.95) \div 4.8 = 6.08 \rightarrow \mathbf{6.5 \text{ kWp}}$$

Extra solar charges the battery for the night. U = usable battery energy, $\eta = 95\%$.

Both sizes round up to the next 0.5 kWp.

STEP 3 · BATTERY

Battery capacity C (kWh)

$$C_{\text{required}} = N \div (\text{DoD} \times \eta)$$

$$15.60 \div (0.80 \times 0.95) = 20.53 \text{ kWh}$$

Pick the smallest AVESTA combination at or above it: any number of LV_32, plus at most one LV_16 and one LV_5. Ties go to fewer units.

$$\rightarrow \text{LV}_{16} + \text{LV}_5 = 21.12 \text{ kWh}$$
$$\text{Usable } U = C \times \text{DoD} = 16.90 \text{ kWh}$$

“Full day” mode uses E instead of N, the original deck's method.

Required kWh	Combination
up to 5.12	1 × LV_5
up to 16	1 × LV_16
up to 21.12	LV_16 + LV_5
up to 32	1 × LV_32
up to 37.12	LV_32 + LV_5
up to 48	LV_32 + LV_16
up to 53.12	LV_32 + LV_16 + LV_5
up to 64	2 × LV_32

STEP 4 • NEW BILL

Grid import after solar (kWh per day)

Equation	Solar only	Solar + battery
Generation $G = P \times Y$	14.40	31.20
Direct use = $\min(G, D)$	12.77	12.77
Surplus = $G - \text{direct}$	1.63 (lost)	18.43
From battery = $\min(\text{surplus} \times \eta, U, N)$	0	15.60
Import / day = $E - \text{direct} - \text{battery}$	15.60	0
Import / month = $\times 30$	468 kWh	0 kWh
New bill = $\text{Bill}(\text{import})$	717 EGP • Tier 5	0 EGP

Exported surplus earns nothing: savings count only solar energy the home uses itself.

STEP 5 · ECONOMICS

System price (EGP)

Inverter kW = smallest standard size $\geq \max(3, P)$: 3, 5, 6, 8, 10, 12, 15, 20 ...

Price = $(P \times 8,000 + \text{Inverter} \times 6,000) \times 1.10 + C \times 13,640$

The 10% for installation & BOS applies to PV and inverter only, not to the battery.

Solar only · 3 kWp, 3 kW inverter

$(24,000 + 18,000) \times 1.10$

46,200

Solar + battery · 6.5 kWp, 8 kW inverter

$(52,000 + 48,000) \times 1.10 + 288,077$

398,077

Savings, payback and backup

Equation	Solar only	Solar + battery
Monthly saving = Bill(M) – Bill(import)	1,283	2,000
Annual saving = monthly × 12	15,397	24,001
Payback = Price ÷ annual saving	3.0 years	16.6 years
Backup hours = U ÷ (E ÷ 24)	none	≈ 14.3 h

Recommend solar + battery when its payback ≤ 1.15 × solar-only payback

Backup hours use the average load; peaks such as air conditioning shorten them.

TIER CLIFF

The kWh just above 650 or 1000 are the most valuable to cut

Value per kWh cut = $(\text{Bill}(M) - \text{Bill}(T)) \div (M - T)$, with $T = 650$ or 1000

4.41

EGP saved per kWh: $(2,000 - 1,113.5) \div (851 - 650)$

2.35

EGP average price today

INPUTS

Default assumptions and their sources

Assumption	Default	Source
Night share	55%	Estimate; needs measured load profiles
PV yield	4.8 kWh/kWp/day	Cairo ≈ 1,750 kWh/kWp per year
Depth of discharge	80%	AVESTA LV datasheet: 6,000 cycles at 80%
Round-trip efficiency	95%	Home Storage deck
Battery price	13,640 EGP/kWh	Deck: €220 × 62 EGP
PV price	8,000 EGP/kWp	Deck
Inverter price	6,000 EGP/kW	Estimate
Installation & BOS	10% of PV + inverter	Deck: 10–15%
Days per month	30	Fixed

All defaults except days per month are editable under “Assumptions & prices” in the tool.